

# COGNITIVE COMPLIANCE & FORENSIC AUDITOR

Multi-Agent Courtroom — Audit Report

|                  |                                      |            |           |
|------------------|--------------------------------------|------------|-----------|
| Transaction ID   | TXN-2026-XB-004821                   | Prediction | HIGH RISK |
| Amount           | \$47,500.00 (INR → USD)              | Confidence | 94%       |
| Timestamp        | 2026-03-15T08:42:11Z                 | Frozen     | True      |
| Run ID           | ea0d68a6-b763-4809-967e-2065d496dc83 | SARs Filed | 2         |
| Report Generated | 2026-03-16T12:07:42.023469+00:00     |            |           |

## 1. AUDITOR FINAL VERDICT

AUDITOR FINAL RULING: TRANSACTION FROZEN

Both jurisdictional experts are in agreement: this transaction is HIGH RISK and NON-COMPLIANT across RBI (India) and EU/AMLD-6 frameworks.

KEY AGGRAVATING FACTORS:

- GNN cycle motif CYCLE-A detected — textbook smurfing pattern.
- Shell company probability 87% (above both RBI and EU thresholds).
- Sanctions hit on intermediary ACC-AE-0077 (UAE).
- Both end-point accounts have PENDING KYC and unverified beneficial owners.
- 94-day and 61-day old accounts — both below the 180-day safe-harbour.

ACTIONS ORDERED:

- Immediately freeze TXN-2026-XB-004821.
- File STR with FIU-IND (India) within 24 hours.
- File SAR with GoAML (EU) within 24 hours.
- Escalate ACC-AE-0077 to the Sanctions Compliance team.
- Place a 90-day monitoring hold on cluster CLUSTER-SHELL-07.

CONFIDENCE: 94%

XAI REPORT: Attached.

## 2. JURISDICTIONAL EXPERT VERDICTS

### India Expert (RBI)

VERDICT: HIGH RISK – NON-COMPLIANT

Under RBI jurisdiction I find three critical breaches:

- [RBI-KYC-2016-§38] Amount \$47,500 exceeds ₹25 lakh EDD threshold. No source-of-funds declaration is on file. BREACH.
- [RBI-FEMA-2013-§6(3)] Two intermediary hops (Singapore, UAE) within 72 hours trigger mandatory FIU-IND reporting. BREACH.
- [RBI-AML-2023-§12] Sender account age = 94 days. Transaction exceeds ₹15 lakh monthly cap for unverified new accounts. BREACH.

Sender KYC status: PENDING. Beneficial owner: UNVERIFIED.

RECOMMENDATION: Freeze transaction. File STR with FIU-IND within 24 hours.

### **EU Expert (AMLD-6 / AI Act)**

VERDICT: HIGH RISK – PARTIAL BREACH

Under EU/AMLD-6 and EU AI Act jurisdiction:

1. [AMLD6-Art3(4)] GNN shell-company probability = 87%, exceeding the 80% threshold. A SAR must be filed with GoAML within 24 hours. BREACH.
2. [EUAIA-2024-Art22] The CCFA system is generating this XAI report, satisfying the explainability obligation. COMPLIANT.

Intermediary account ACC-AE-0077 (UAE) has a SANCTIONS HIT and PEP flag in KYC database — this is an aggravating factor under AMLD-6 Art 18.

RECOMMENDATION: Freeze transaction. Mandatory SAR via GoAML. Escalate ACC-AE-0077 to sanctions compliance team.

### 3. REGULATORY BREACH SUMMARY

| Rule ID             | Status      | Reason                                                                    | Penalty Cap                                           |
|---------------------|-------------|---------------------------------------------------------------------------|-------------------------------------------------------|
| RBI-KYC-2016-§38    | BREACH ✗    | Amount ≈\$47,500 (>\$30k threshold). No source-of-funds document updated. | Up to 1% of global annual turnover or ₹1 crore.       |
| RBI-FEMA-2013-§6(3) | BREACH ✗    | 2 intermediate hops (Singapore, UAE) within 72 hrs.                       | Up to 4% of global annual turnover or ₹1 crore.       |
| RBI-AML-2023-§12    | BREACH ✗    | Sender account age = 94 days; transaction amount exceeds cap.             | Up to 4% of global annual turnover or ₹1 crore.       |
| EUAIA-2024-Art22    | COMPLIANT ✓ | CCFA system generates XAI report per this rule — compliant.               | Up to €10 million or 2% of worldwide annual turnover. |
| AMLD6-Art3(4)       | BREACH ✗    | Shell company probability = 87%. SAR not yet filed.                       | Up to €10 million or 2% of worldwide annual turnover. |

### 4. SHAP FEATURE ATTRIBUTIONS (XAI)

Base value (population average risk): 0.350 | Final model output: 0.940 | Model: Pytorch Geometric GAT v2.1.0-mock

| Feature                                             | SHAP Value | LIME Weight | Direction | Raw Value                              |
|-----------------------------------------------------|------------|-------------|-----------|----------------------------------------|
| Circular money flow (smurfing cycle) detected       | +0.4230    | 0.4441      | ▲ RISK    | [{'cycle_id': 'CYCLE-2026-03-16-001'}] |
| Overall graph anomaly score from GNN                | +0.3458    | 0.3631      | ▲ RISK    | 0.91                                   |
| Probability that sender/receiver is a shell company | +0.2784    | 0.2923      | ▲ RISK    | 0.87                                   |
| Number of known bad actors in same cluster          | +0.2324    | 0.2440      | ▲ RISK    | 3                                      |
| Transaction velocity spike (3× 48-hr baseline)      | +0.2125    | 0.2231      | ▲ RISK    | True                                   |
| Suspicious intermediary routing hops                | +0.1738    | 0.1825      | ▲ RISK    | [{'account_id': 'ACC-2026-03-16-002'}] |
| Amounts vary slightly to evade filters (smurfing)   | +0.1520    | 0.1596      | ▲ RISK    | True                                   |
| Sender account is very new (< 180 days)             | +0.1296    | 0.1361      | ▲ RISK    | True                                   |
| Receiver account is very new (< 180 days)           | +0.1088    | 0.1142      | ▲ RISK    | True                                   |
| Device fingerprint matches known good device        | -0.0610    | 0.0640      | ▼ SAFE    | False                                  |
| Sender provided a stated purpose for transfer       | -0.0500    | 0.0525      | ▼ SAFE    | False                                  |
| Sender bank is a recognised Tier-1 institution      | -0.0300    | 0.0315      | ▼ SAFE    | False                                  |

### 5. LIME LOCAL EXPLANATION

LIME local approximation (neighbourhood size=500 perturbations):

The GNN classifier assigned HIGH RISK to transaction TXN-2026-XB-004821 primarily because similar transactions in its neighbourhood (±15% amount, ±30 days) with cycle motifs AND sender account age <120 days were classified HIGH RISK in 98.2% of cases. The most influential boundary feature is the presence of a confirmed cycle motif (CYCLE-A): removing it reduces the risk score from 0.91 to 0.43. The velocity spike (3× 48h baseline) adds an independent +0.18 to the risk logit. Beneficiary account age <90 days adds +0.14. These three features together account for 79% of the decision.

### 6. SAR / STR FILING RECEIPTS

| Jurisdiction | Tracking ID               | Filed At                         | Review Days |
|--------------|---------------------------|----------------------------------|-------------|
| RBI          | STR-FIUIND-20260316-19238 | 2026-03-16T12:07:42.021179+00:00 | 3           |
| EU           | SAR-GOAML-20260316-82419  | 2026-03-16T12:07:42.021179+00:00 | 3           |

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